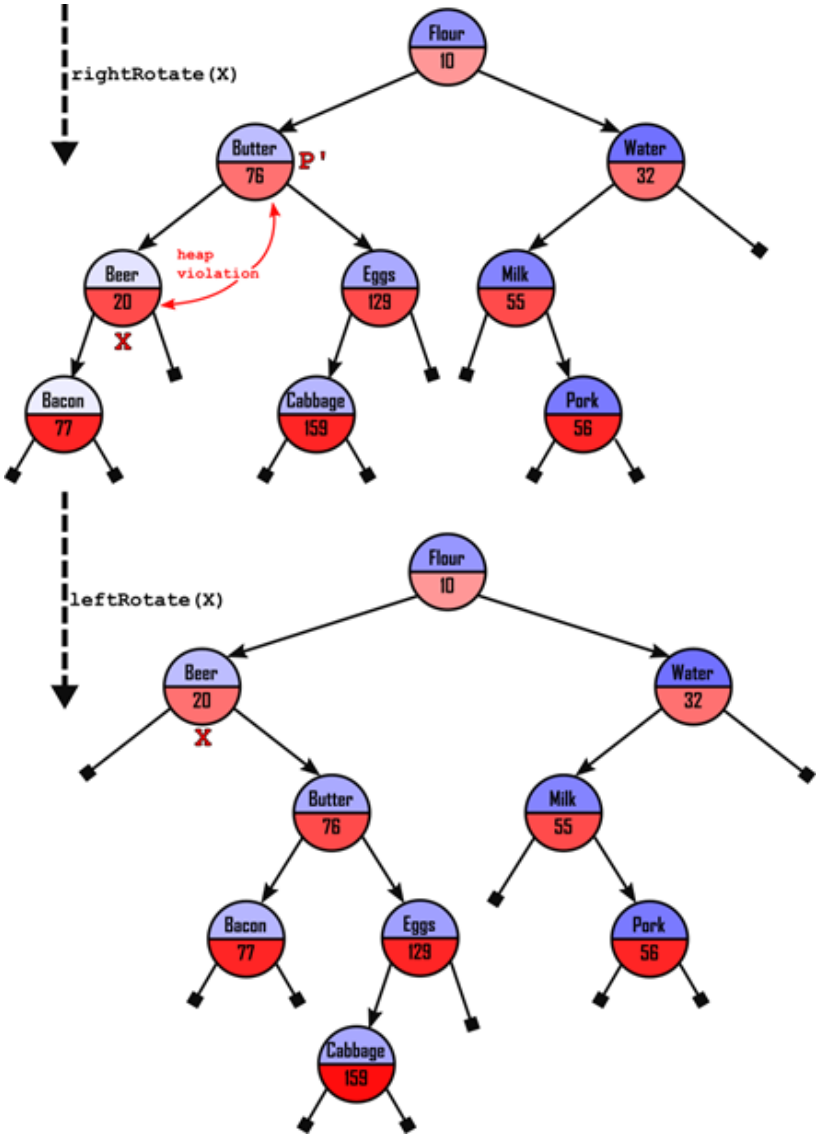



**Figure 3.9 Insertion (continued). To fix heap's invariants, we needed to perform a right rotation on the treap at the end of figure 3.8; this action bubbles up one level the new node, and the result is shown at the top of this figure. Heap's invariants are still violated, so we need to perform a further rotation (left, this one).**

## 3.3.5 Delete

Deleting a key from the treap is a conceptually simpler operation, although it requires a completely different approach with respect